

Task Description:

Install docker on EC2 and explore the docker commands (docker images, containers, volumes, network)

Output

```
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-9-188:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-11-26 07:57:56 UTC; 45s ago
 TriggeredBy: ● docker.socket
    Docs: https://docs.docker.com
   Main PID: 3423 (dockerd)
      Tasks: 8
   Memory: 103.3M (peak: 103.8M)
      CPU: 393ms
   CGroup: /system.slice/docker.service
           └─3423 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.375776867Z" level=info msg="Restoring containers: start."
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.411289976Z" level=info msg="Deleting nftables IPv4 rules" error="exit status 1"
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.417471672Z" level=info msg="Deleting nftables IPv6 rules" error="exit status 1"
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.705720523Z" level=info msg="Loading containers: done."
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.757065040Z" level=info msg="Docker daemon" commit=4612690 containerd-snapshotter=true storage-driver=
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.757338626Z" level=info msg="Initializing buildkit"
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.812463444Z" level=info msg="Completed buildkit initialization"
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.820852385Z" level=info msg="Daemon has completed initialization"
Nov 26 07:57:56 ip-172-31-9-188 dockerd[3423]: time="2025-11-26T07:57:56.821132411Z" level=info msg="API listen on /run/docker.sock"
Nov 26 07:57:56 ip-172-31-9-188 systemd[1]: Started docker.service - Docker Application Container Engine.
```

```
ubuntu@ip-172-31-9-188:~$ sudo usermod -aG docker $USER
ubuntu@ip-172-31-9-188:~$ newgrp docker
ubuntu@ip-172-31-9-188:~$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
ubuntu@ip-172-31-9-188:~$ docker --version
Docker version 29.0.4, build 3247a5a
ubuntu@ip-172-31-9-188:~$ docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
0e4bc2bd6656: Pull complete
de57a609c9d5: Pull complete
53d743880af4: Pull complete
b5feb73171bf: Pull complete
77fa2eb06317: Pull complete
108ab8292820: Pull complete
192e2451f875: Pull complete
Digest: sha256:553f64aecdc31b5bf944521731cd70e35da4faed96b2b7548a3d8e2598c52a42
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-9-188:~$
```

```
77fa2eb06317: Pull complete
108ab8292820: Pull complete
192e2451f875: Pull complete
Digest: sha256:553f64aecdc31b5bf944521731cd70e35da4faed96b2b7548a3d8e2598c52a42
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-9-188:~$ docker image
Usage:  docker image COMMAND

Manage images

Commands:
  build      Build an image from a Dockerfile
  history    Show the history of an image
  import     Import the contents from a tarball to create a filesystem image
  inspect    Display detailed information on one or more images
  load       Load an image from a tar archive or STDIN
  ls         List images
  prune      Remove unused images
  pull       Download an image from a registry
  push       Upload an image to a registry
  rm         Remove one or more images
  save       Save one or more images to a tar archive (streamed to STDOUT by default)
  tag        Create a tag TARGET_IMAGE that refers to SOURCE_IMAGE

Run 'docker image COMMAND --help' for more information on a command.
ubuntu@ip-172-31-9-188:~$ docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	553f64aecdc3	225MB	59.8MB	

Info → In Use

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```
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Manage images

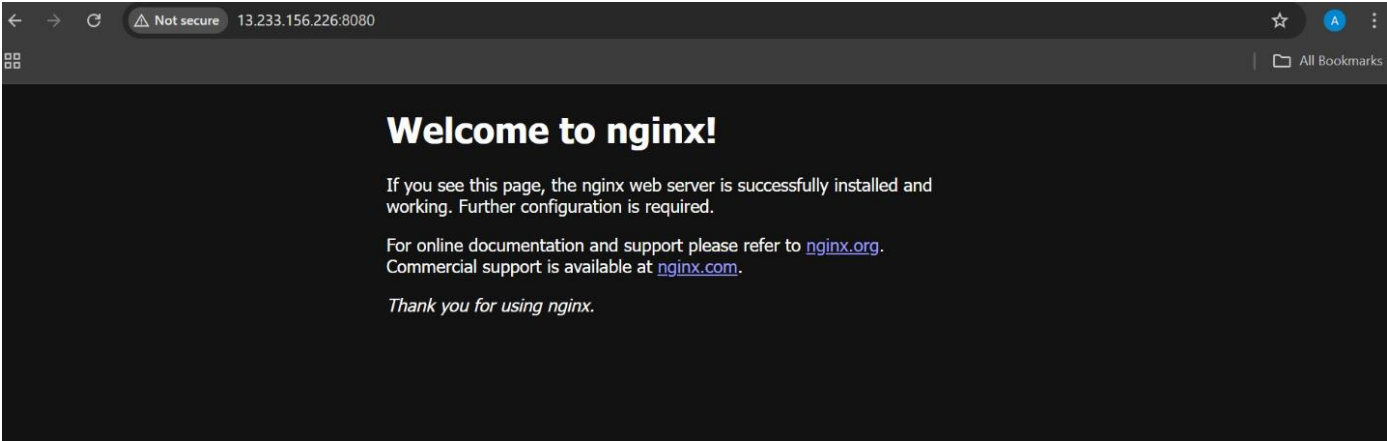
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```
ubuntu@ip-172-31-9-188:~$ docker run -d -p 8080:80 nginx
7915622d2d2ab93e054da92a8ee07cf958d2dbae20d4ec61f2a5fetc057799201
```

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```
ubuntu@ip-172-31-9-188:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
7915622d2ab   nginx    "/docker-entrypoint..." About a minute ago Up About a minute 0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nifty_wilson

ubuntu@ip-172-31-9-188:~$ docker volume create myvol
myvol

ubuntu@ip-172-31-9-188:~$ docker volume ls
DRIVER      VOLUME NAME
local       myvol

ubuntu@ip-172-31-9-188:~$ docker network ls
NETWORK ID    NAME        DRIVER    SCOPE
2fe54762645f  bridge     bridge    local
f3b46335a6ff  host       host      local
5026d9b6f12a  none      null      local

ubuntu@ip-172-31-9-188:~$
```